

Climate Impact Analysis of Bat Lake Salamanders (*A. maculatum*)

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This report describes an analytical pipeline forecasting a northern population of yellow spotted salamanders over the remainder of the 21st century. The analysis exists as part of a larger research paper (Kaufman et al., 20XX), and much of the text, figures, and tables in this report were taken directly from there. All code and data used to complete this sub-analysis are available at <https://github.com/lepasserin/a-maculatum-climate-impact>.

1 Abstract (from pre-print)

Temperate regions experience pronounced interannual climatic variability, providing the opportunity to understand the drivers of temporal variation in survival, and to model future impacts of environmental change. Temperate amphibians are sensitive to climatic variation and are experiencing dramatic declines in range size and abundance under global environmental change. We developed a priori hypotheses for climate effects on amphibian survival, then combine fifteen years of mark-recapture data from a population of spotted salamanders (*Ambystoma maculatum*) with Bayesian hierarchical modelling to test the association between environmental variables and annual survival. We found that greater temperatures during the spring inbound breeding migration led to a 50% decline in annual survival, while greater precipitation during the outbound migration was associated with a survival increase of 30%. Evaporative water loss may be the major driver of survival variation during the spring migration.

We then projected estimates of adult female survival to the year 2100 under three climate change scenarios while incorporating the estimated effects of environmental variables on adult survival and assuming that other population parameters would be unchanged. In the mildest scenario, the projected increase in spring precipitation were able to compensate for their warmer temperatures, resulting in an approximately stable population size. Faced with more extreme future climates, the population declines to over half of its initial size. Our study draws urgent attention to the migratory dynamics of temperate, spring breeding amphibians faced with a changing climate.

2 Methods: Projecting Survivorship & Abundance under Climate Change (from pre-print)

We paired results from the posthoc breeding migration model with climate forecasting data to project estimates of annual survivorship for the Bat Lake Spotted Salamander population through the remainder of the 21st century. First, we forecast the inbound temperature and outbound precipitation covariates from the breeding migration model using climate data from global circulation models (GCMs), under three climate scenarios. These forecasts were then used to estimate future adult female annual survival probabilities. These survivorship estimates were also embedded into a matrix model to project the potential impacts of these changing climatic indices on abundance, in the absence of other effects.

Estimates of mean temperature, minimum temperature and total precipitation were required for each day between March–May for the years 2005–2100 to calculate the migration model’s environmental covariates. These data were sourced from the Coupled Model Intercomparison Project Phase 6 (CMIP6) dataset, which provides daily weather predictions that have been statistically downscaled to spatial resolutions of 10 km² (Eyring et al. 2016). We used the closest available grid point to the Algonquin East Gate Climate Station (45° 32’ N, 78° 17’ W), approximately 2.2 km away. For each climate variable, the ensemble mean of the 26 available GCMs was used. Three such datasets were compiled, one for each of three Shared

Socio-economic Pathways (SSPs) representing conservative (SSP1), moderate (SSP2) and extreme (SSP5) fossil-fueled development respectively (Riahi et al. 2017).

We applied two statistical bias corrections to the GCM data, in accordance with standard practice for climate impact analysis (Sharma et al. 2007; Hagemann et al. 2011). For mean and minimum temperature, additive linear scaling was used to correct for long-run bias in mean and minimum temperature data, following the procedure described in Hempel et al. (2013). For total precipitation, local intensity scaling was used to correct the intensity and frequency of rainy days, applied as described in Smitha et al. (2018). These procedures were applied to the GCM data over a 20-year period (2005-2024) where their results could be directly compared to the Algonquin East Gate Climate Station’s historical records. Once suitable correction factors had been identified, these corrections were applied to future GCM data (see Appendix A for explicit correction equations and a visual comparison of raw versus bias-corrected simulated data).

Once bias-corrected, GCM data was used to predict the salamander breeding season start and end dates, enabling the calculation of the specific environmental covariates used in the CJS model, for each year through to 2100. For a given year, the breeding season start date was estimated using that year’s mean April temperature (leveraging its strong, negative correlation with Bat Lake ice out date; Supplement B), and was set to end 22 days later, corresponding to the average breeding season length observed during the BLISS study period (2008 to 2022). Estimated intervals were validated by comparing hindcasted and observed environmental indices over this same period (Supplement B). Using these indices, annual survivorship estimates were calculated for each year from 2005 to 2100 using the breeding migration model. This process was repeated for each of the three climate scenarios. Estimated survival probabilities were validated by comparing hindcasted estimates and fitted estimates based on observed data from 2008 to 2022 (Supplement B).

Finally, we used these estimated survival probabilities within a simple matrix model to project long-term population trajectories under different future climate scenarios, and in the absence of climate-driven effects on other population vital rates. Our chosen population model is a

deterministic, female-only, age-structured matrix model where the final class represents all reproductively mature adults (Smith and Trout 1994). We chose this model over a simpler one to account for the time lag between changes in adult survivorship and their effects on future adult recruitment. Since female spotted salamanders mature at around 6 years old (Boyle, 2012), this resulted in a seven-staged discrete growth model. We sourced fecundity estimates (i.e. the number of year-old females per adult female alive in the previous year) for the Bat Lake population from Moldowan (2023). As annual juvenile survivorship from ages 1 to 6 was unknown, these values were set such that the resulting model had a finite rate of increase (λ) of 1 under baseline conditions (Congdon et al. 1994). We considered the baseline conditions to be the average values of the observed environmental indices over the BLISS study period (see Supplement C for model parameterization). Projections were initialized in the year 2019 at the base model’s stable age distribution with 1840 adult females. We tested the sensitivity of our projections to changes in initial abundance, starting year, and all non-focal vital rates (fecundity, juvenile survival, age of maturity, holding $\lambda = 1$) finding that such variables had little effect on our results, within reasonable searching ranges Supplement C.

3 Results: Projecting Survivorship & Abundance under Climate Change (from pre-print)

Supplement

A Statistical Bias Corrections

The daily projections of mean temperature, minimum temperature, and total precipitation used in this study (hereafter referred to as “climate projections”) were sourced originally from the Coupled Model Intercomparison Project Phase 6 (CMIP6) dataset, which provides daily estimates of these climate variables under various socio-political scenarios (called “Shared Socioeconomic Pathways”, often abbreviated as SSPs). Since GCM output is coarse-grained, Environment and Climate Change Canada provides downscaled estimates with higher spatial resolution [10 km²; Eyring et al. (2016)], for each of three SSPs (SSP1, SSP2, and SSP5 ; see (Riahi et al. 2017) for detailed definitions). These are more appropriate for the analysis of highly localized populations such as the Bat Lake spotted salamander population. Even with downscaling, however, climate projections are often still systematically biased (Sharma et al. 2007; Hagemann et al. 2011); while observed trends are often accurately reproduced, estimates may deviate substantially from observations in their absolute values and/or interannual variations.

We implemented two bias correction methods to correct specific, systematic deviations between observed and simulated climate metrics. In both cases, these methods were first applied over a historical period (2005 to 2024) where observed daily climate data was available near our study site (measurements taken from Algonquin East Gate Climate Station), to determine suitable correction factors. Then, these correction factors were applied to correct bias in future projections (through to 2100), under the assumption that systematic bias in the past will manifest itself in the same way in the future. We note here that projections from all three SSPs considered were corrected using the same coefficients.

A.1 Correcting Temperature Forecasts

For both mean and minimum daily temperature, downscaled estimates showed similar trends to observed measurements over the entire historical period, but with consistently higher monthly averages. Simulated temperatures also exhibited smaller day-to-day variation compared to observed trends, likely because our simulated estimates are based on the ensemble mean of 26 different GCMs, which artificially deflates variance. Methods that correct the variance and the mean in simulated temperature data simultaneously exist, but have been shown to impact the climate change signal in certain datasets (Hagemann et al. 2011). Given how temperature projections are used in further analyses, we opted to only correct the systematic bias in the average, preserving the climate change signal most faithfully. To this end we applied additive linear scaling following Hempel et al. (2013), implemented as described below.

Let i represent the March of a given year between 2005 and 2024. Then, let T_{ij}^{OBS} represent the observed temperature (mean or minimum) of the j^{th} day in that month respectively. T_{ij}^{SIM} and T_{ij}^{ADJ} are defined similarly, for simulated (SIM) and bias-corrected (ADJ) daily values. We calculate the linear scaling constant C as follows:

$$C = \left(\sum_{i=2005}^{2024} T_i^{\text{OBS}} - T_i^{\text{SIM}} \right) / 20 \quad (1)$$

where T_i^{OBS} and T_i^{SIM} represent the average temperature of the i^{th} March from 2005 to 2024, for observed and simulated data respectively. C can then be used to scale future daily temperature simulations like so:

$$T_{i,j}^{\text{ADJ}} = C + T_{i,j}^{\text{SIM}} \quad (2)$$

Notice that C is calculated over the entire period, and not per-year. This allows us to apply the correction shown in Equation 2 to every March in the forecasted simulated data. This procedure can then be repeated for April and May, mean and minimum temperatures separately,

resulting in the estimation of 6 linear scaling constants to correct GCM-derived temperature forecasts.

A.2 Correcting Precipitation Forecasts

For daily total precipitation, downscaled estimates had similar monthly and seasonal averages as observed measurements, but grossly underestimated the day-to-day variance in precipitation, effectively predicting a drizzle each day. This resulted in an underestimated number of rainy days, and an underestimated intensity of wet days. Since the precipitation main effect used in our analysis is sensitive to the number of rainy days (see main text), this drizzle bias required correction. To this end we used local intensity scaling following Smitha et al. (2018), described below.

Let i represent the March of a given year between 2005 and 2024. Then, let P_{ij}^{OBS} represent the observed precipitation (total) on the j^{th} day in that month respectively. P_{ij}^{SIM} and P_{ij}^{ADJ} are defined similarly, for simulated (SIM) and bias-corrected (ADJ) daily values. First, a fixed threshold t is determined such that the average number of wet days with simulated precipitation totals below t is equal to the average number of observed dry days. This constant is used to correct for the number of dry days in the simulated data. Then, we calculate the observed average precipitation among wet days (D_1), and the average precipitation among simulated days above t (D_2). Their ratio, $D = \frac{D_1}{D_2}$, is used to correct for the expected intensity of wet days in the simulated data. These constants are then applied to the simulated precipitation data as follows:

$$P_{i,j}^{\text{ADJ}} = \begin{cases} 0, & \text{if } P_{i,j}^{\text{SIM}} < t \\ (P_{i,j}^{\text{SIM}} - t) \times D, & \text{otherwise} \end{cases} \quad (3)$$

Similarly to temperature, three sets of (D, t) pairs were fit, for each month in our dataset.

A.3 Bias Correction Results & Diagnostics

All fitted correction factors are shown in Table A1. Monthly biases in simulated temperatures were more extreme earlier in the season, and for minimum vs. average daily temperature estimates. Simulated precipitation days were deemed dry when their total precipitation did not exceed around 2mm, equivalent to a light shower. Simulated wet days saw their total precipitation amount inflated by a factor of 4 to 7. By construction, the biases we targeted for each climate variable were completely removed when values are averaged over the entire period. However, we also found that, for any specific month in the historical data, raw GCM data was more likely to deviate from observed estimates than bias-corrected GCM data.

Figure A1 compares observed, simulated, and bias corrected daily weather estimates for an arbitrarily chosen year in the historical period (2010). This is shown to illustrate how the correction algorithms modify the simulated data; the effects were qualitatively similar for any chosen year within the historical period (2005 to 2024). Since all temperature bias correction coefficients are negative, the simulated trend is pulled downwards. For precipitation, days receiving under ~2mm of precipitation are deemed dry, and wet days see their precipitation estimates inflated, creating daily variability that is similar to the observed trend.

Table A1: Fitted correction factors applied to simulated climate projections.

Correction Factor	Value	Correction Factor	Value
March mean temperature offset (°C)	-1.54834	March dry day threshold (mm)	2.11449
April mean temperature offset (°C)	-1.38055	April dry day threshold (mm)	2.16254
May mean temperature offset (°C)	-0.57552	May dry day threshold (mm)	2.57951
March minimum temperature offset (°C)	-2.91662	March intensity coefficient (unitless)	4.03366
April minimum temperature offset (°C)	-1.90356	April intensity coefficient (unitless)	5.97328
May minimum temperature offset (°C)	-1.66153	May intensity coefficient (unitless)	6.87419

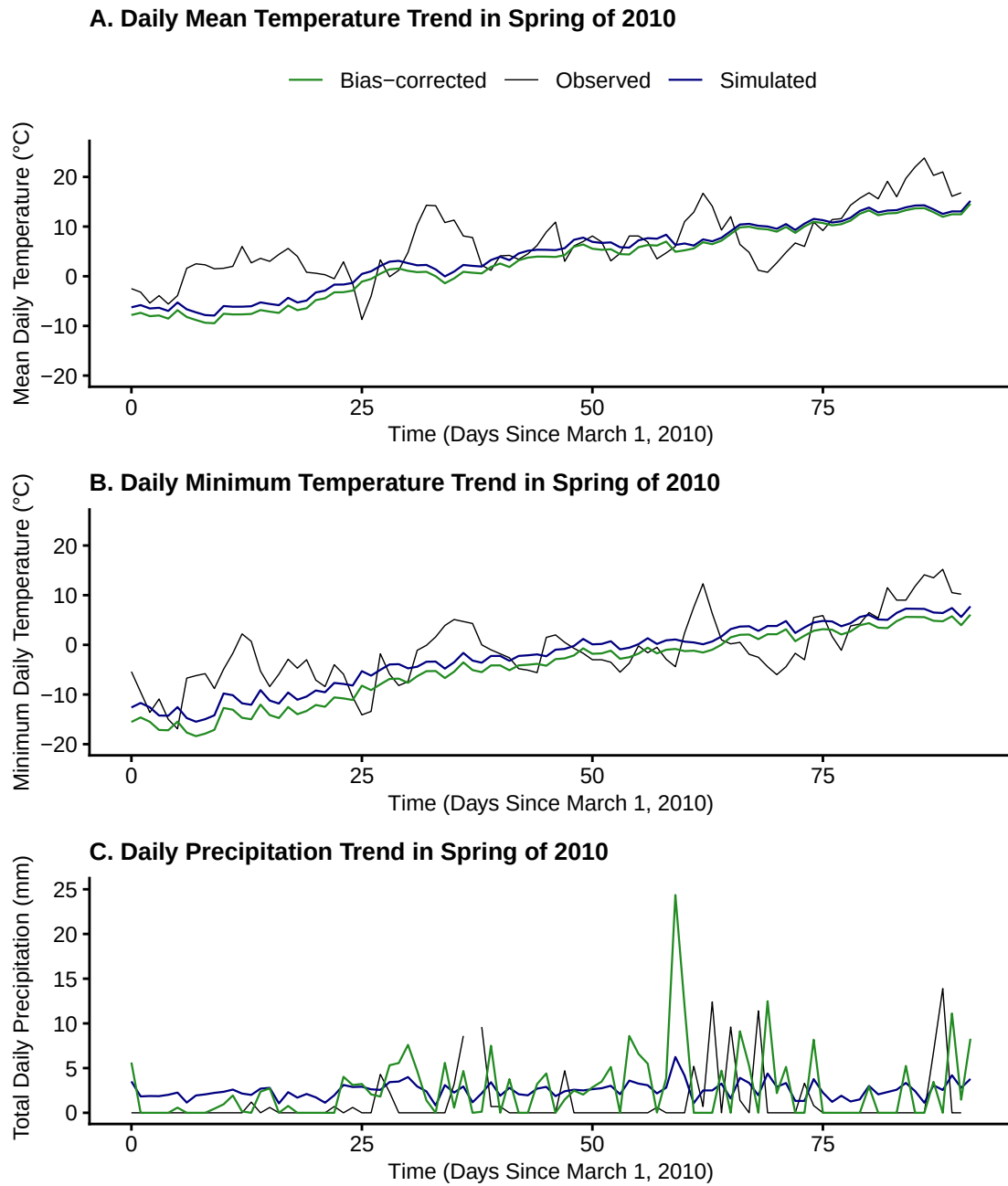


Figure A1: Time series of observed (black), simulated (blue), and bias-corrected (green) daily weather estimates from March 1 to May 31, 2010. Panel A displays mean daily temperature, panel B displays minimum daily temperature, and panel C displays total daily precipitation.

B Forecasting Environmental Indices & Survival Probabilities

Bias-corrected daily weather forecasts were used to calculate four key measures, for each year from 2005 to 2100. First, average mean daily temperatures in April were used to predict the breeding season start date (during the BLISS study period, these measures were strongly correlated; $|\rho| = 0.5$). Breeding season end date was calculated directly from the start date, by adding the average breeding season length observed during the BLISS study period. Using the forecasted breeding season interval, we calculated the temperature and precipitation indices as they were calculated in the posthoc breeding migration model (see main text for their formulas). This series of predictions and calculations was made for each climate scenario separately.

Figure B1 displays the resulting time series of estimates for the breeding season start date and both climate indices. These graphs are also overlayed with the observed values during the BLISS study period. Average index hindcasts over this period were very similar to observed averages, for all climate scenarios. However, observed temperature indices displayed more annual variability than hindcasted ones.

Forecasted climate indices (panel B and C in Figure B1) were used to estimate adult female survival probabilities over the century, using the posthoc breeding migration model fit in the main text. The survival probability at year t ($S_a(t)$) was derived from the temperature (Temp) and precipitation (Precip) indices estimated for that year using the following formula:

$$S_a(t) = \text{logit}^{-1} [\beta_0 + \beta_1 \text{Temp}(t) + \beta_2 \text{Precip}(t)] \quad (4)$$

where β_1 , β_2 , and β_3 were fit using the breeding migration model. The resulting survival forecast for each climate scenario is displayed in Figure B2.

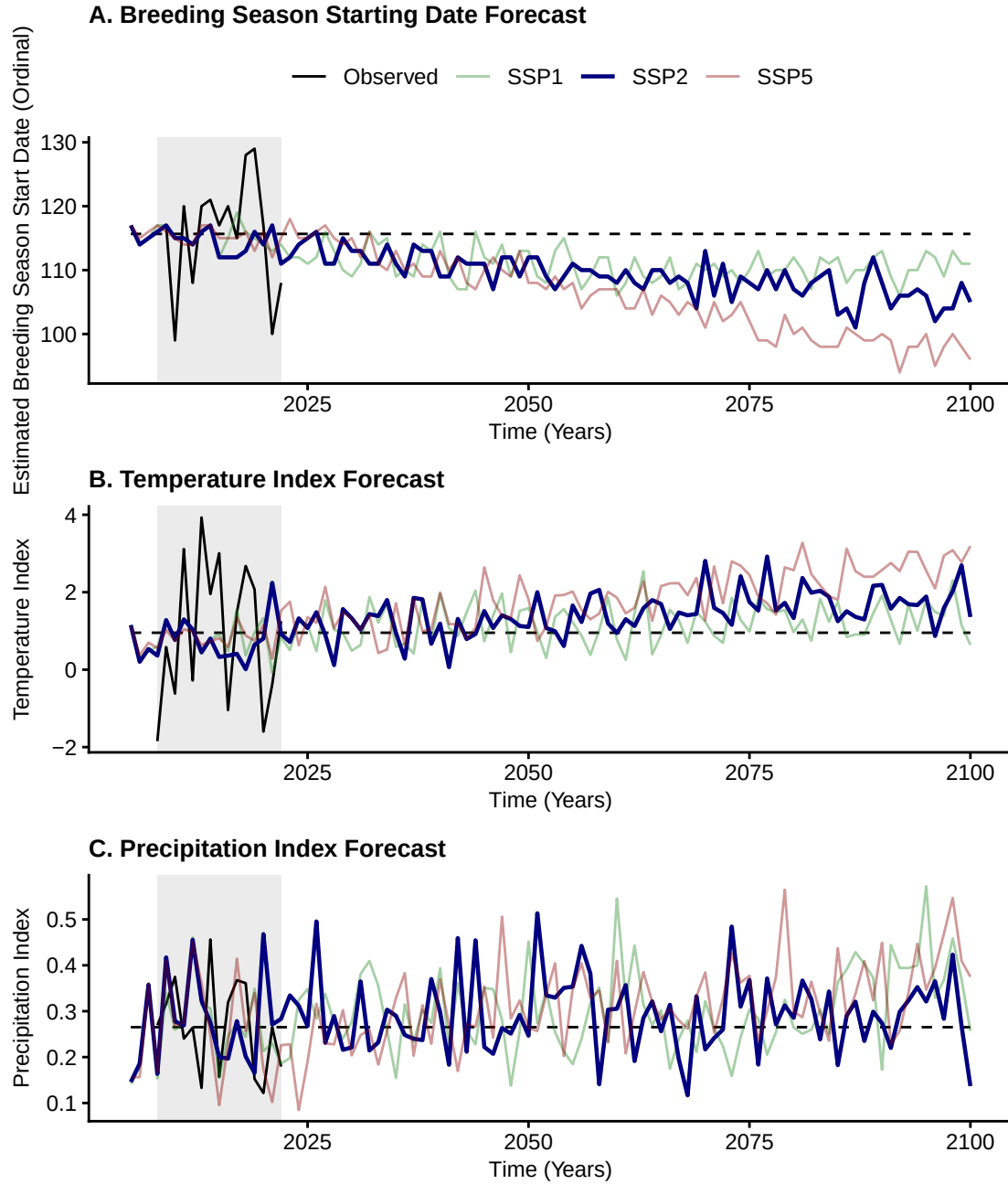


Figure B1: Time series of observed (black), and forecasted (coloured) indices, from 2005 to 2100. Panels A, B, and C display breeding season start date, temperature, and precipitation indices respectively. The grey shaded areas represent the BLISS study period (2008 to 2022), and the horizontal dashed lines indicate the average observed index values during this period. The climate scenario labelled ‘SSP2’ is emphasized for visual contrast.

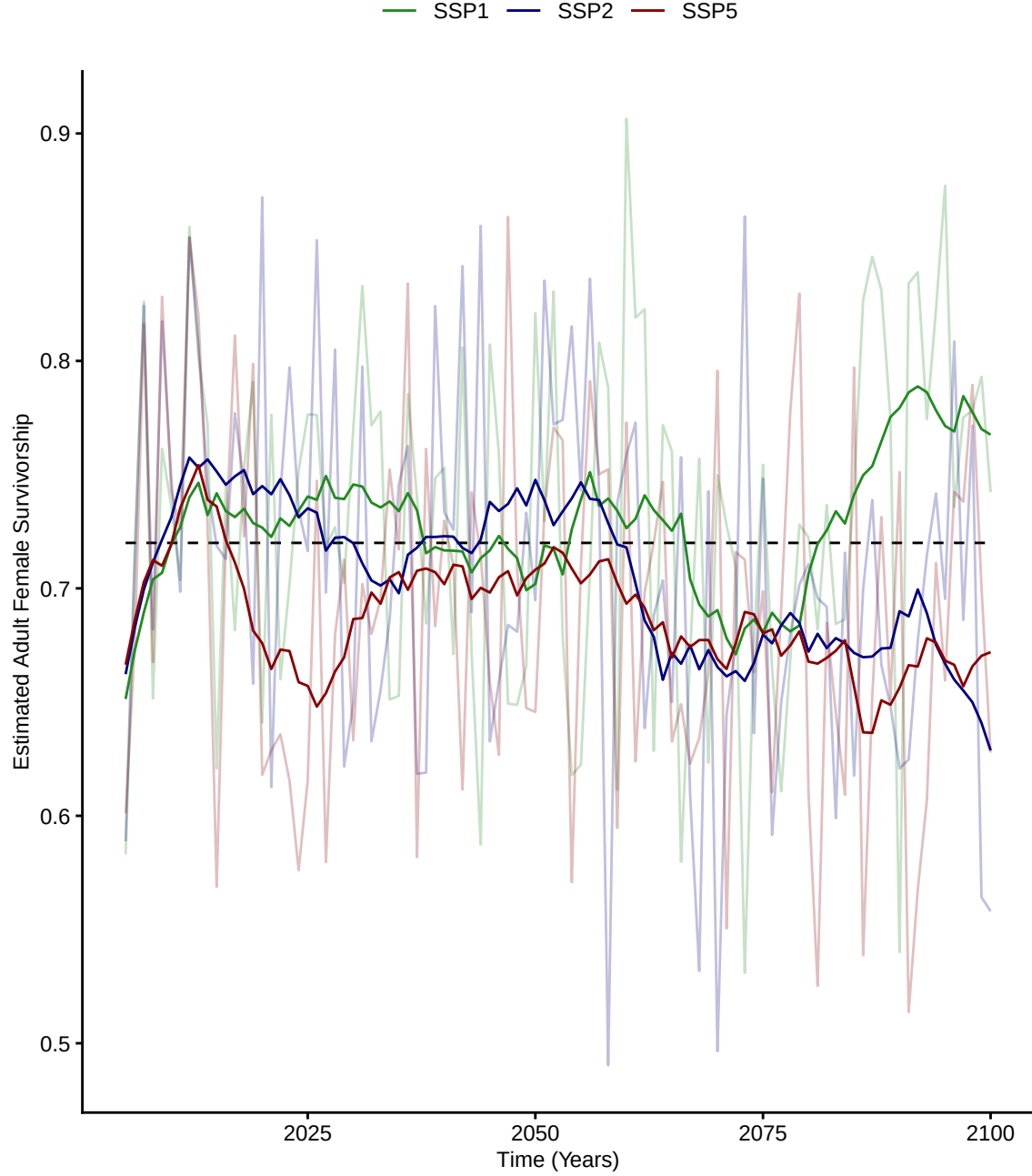


Figure B2: Time series of estimated adult female survivorship (S_a) from 2005 to 2100, based on environmental data derived from global climate model projections. Three climate scenario (SSP1, SSP2, and SSP3) were considered, and their trends are colored accordingly. The horizontal dashed line indicates the adult female survivorship estimated at the average environmental values observed during the BLISS study period (2008 to 2022). Both smoothed (bold) and raw (faded) projections are shown. Smoothed trends are emphasized to better visualize differences between the climate scenarios.

C Population Growth Model Specification

We used a simple, deterministic matrix model to forecast the female Bat Lake spotted salamander population through time. Let $\mathbf{N}_t$ be a vector representing the population of salamanders, binned into age groups, in year t . Then, the population in the following year ($\mathbf{N}_{t+1}$) is calculated as follows:

$$\mathbf{N}_{t+1} = \mathbf{N}_t \cdot \mathbf{L}_t \quad (5)$$

where $\mathbf{L}_t$ is the population projection matrix in year t , defined to be:

$$\mathbf{L}_t = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & f \\ S_m & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & S_m & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & S_m & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & S_m & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & S_m & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & S_m & S_a(t) \end{bmatrix} \quad (6)$$

Population growth in this model is governed primarily by the annual adult female survivorship term $S_a(t)$, which varies from year to year according to those environmental characteristics which correlated with survivorship in the CJS survivorship model fit in the main paper (Equation 4). However, population growth in this model also depends on two other life-history parameters (f and S_m), which were assumed to be constant over time. f represents the average number of female offspring that will survive a full year that are produced by a single mature female. This was set to 0.705, derived from Moldowan, 2023. S_m represents subadult survivorship, and an estimate for this value could not be found in the literature. As such, S_m was numerically optimized such that the resulting model had a finite rate of increase λ near 1, when $S_a(t) = \text{logit}^{-1}(\beta_0) \approx 0.72$ (i.e., adult female survivorship assuming study-average

temperature and precipitation indices). We thought this decision sensible since the population did not show signs of increase or decrease during the BLISS study period (2008 to 2022). This baseline survivorship was also used to derive the stable-age distribution of the population, using the `eigen.analysis` function from the `popbio` package in R. Projections were initialized in the year 2019 at the base model’s stable age distribution with 1840 starting adult females.

Given the uncertainty in some of our modelling decisions, we tested the robustness of our population projections by varying the following elements of our model and comparing the output with the base run:

- Different combinations of the non-focal parameters p and S_m (holding $\lambda \approx 1$ in each case)
- Different age-at-maturity (checked from 5 to 9)
- Earlier starting year (checked from 2000 to 2025).
- Different starting abundance (checked from 1000 to 2000).

These tests revealed that our population projection results were similar under other reasonable life-history assumptions and projection initializations. In particular, the relative trends in abundance across different climate scenarios were qualitatively preserved in each case. Therefore, the only effective assumptions being made when translating survivorship forecasts to abundance forecasts is that $\lambda \approx 1$, and that the population is at stable-age distribution.

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